



P31

Fail-Safe & Recovery: Swarm Quorum Abort, Behavioral-Drift Baseline Rollback, and Cascade Containment as Extensions of the P3 Governance Layer

故障安全与恢复：中止漂移回滚级联遏制

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Working Title (EN): Fail-Safe & Recovery: Swarm Quorum Abort, Behavioral-Drift Baseline Rollback, and Cascade Containment as Extensions of the P3 Governance Layer

Consensus governance paper P31 • recovery-layer extension of v7.5.0 P3 "Causal Effect of Fault-Breaker Governance on Completion Rate" • sister paper to P16 (stop decision) and P26 (weighted BFT) • UDOS Reasoning Engine v7.6.0 • Fang Wenxin • 2026-09-21

One sentence first: v7.5.0 P3 already can "recover if curable, isolate and reassign if not, and honestly fail via kill-switch if utterly impossible"; but its "failure stop" is **single-node, rule-triggered**, lacking three things—**quorum abort** when the whole swarm is led astray together (not decided by one circuit breaker), **automatic baseline rollback** when system behavior quietly drifts (not waiting for fault injection to react), and **cascade containment** when faults spread across nodes. This paper maps these three things from Pillar 3 of the AI SAFE² framework onto P3's governance layer as a **design extension proposal**, with all benefits [RESULT NEEDED] .

Evidence scope (stated only once in the whole paper): All UDOS-side numbers go back to v7.5.0 P3 (reports7/matrix_scale_demo.json, evidence grade cpu-proto; the governance trio and responsibility governance are this-repo designs). The external dialogue object is **AI SAFE² governance framework v3.0 (2026-05-18) Pillar 3 Fail-Safe & Recovery** of the Cyber Strategy Institute; its existence and the F3.3 / F3.4 points have been verified in the v7.6.0 evidence ledger (official-website scope); specific thresholds are that framework's claims, not reproduced in UDOS. The existing numbers of three-level circuit breaking / snapshot rollback / reassignment / kill-switch are cpu-proto single-seed pilots. Verification date 2026-09-21.

Abstract

v7.5.0 P3 factorizes the fault governance of the multi-agent matrix into three independently switchable mechanism groups—three-level circuit breaking (AgentBreaker / sub-matrix group breaker / global kill-switch), snapshot rollback (hash-chain Trace truncation), reassignment (in-flight work orders handed to healthy nodes)—plus a responsibility governance group (unique Owner, hash-chain Trace, explicit Stop Condition), and with the discipline of a pre-registered full-factorial RCT freezes the primary metric (completion rate accepted/total) and guardrails (rework, messages, mis-isolation rate, Trace break count, kill-switch mis-trigger rate). In the single-seed pilot (24 work orders, 9 specialists, 7 QA including 2 Byzantine): no faults 24/24 completion, 0 rework; after injecting crash / drop_context / byzantine three fault types still 24/24 completion, all 3 faulty nodes isolated, 5 reworks; in the extreme "all-domain faults + zero retry budget" scenario it honestly triggers kill-switch and reports non-completion rather than faking success (cpu-proto).

P3's governance is already closed-loop, but its "recovery" still has three structural gaps, precisely corresponding to the three things systematized by the external AI SAFE² framework Pillar 3 (Fail-Safe & Recovery, 2026-05). First, **Swarm Quorum Abort (F3.3)**: P3's kill-switch is a global single-point rule, and when the whole swarm is induced by the same class of error and "all circuit breakers look normal," rule thresholds alone cannot recognize it; Pillar 3's approach is to **coordinate shutdown when over 25% of**

Agents become abnormal, and use **cryptographic attestation weighting** to resist Sybil (i.e. prevent a few identity-spoofing nodes from faking "all is well"). Second, **Behavioral Drift Baseline & Rollback (F3.4)**: P3 only reacts to "fault injection" and has no defense against "system behavior quietly drifting without injected faults"; Pillar 3's approach is to build a baseline for normal behavior and automatically roll back to the last known healthy snapshot when drift exceeds a threshold. Third, **cascade containment**: P3's circuit breaking is node-by-node, and a fault may "jump" from an isolated node to its downstream through the handoff chain; cascade containment requires, while isolating the faulty node, suppressing its influence from spreading to the neighborhood along the Trace path.

This paper maps these three things onto UDOS's existing governance layer: Swarm quorum abort shares a source with the committee mechanism of P16/P26 (using the majority judgment of a group of healthy Agents to replace the single-point ignition rule); drift baseline rollback reuses P3's existing hash-chain Trace snapshots; cascade containment reuses P10's TransferBundle handoff chain. This paper explicitly states: **these three are design extensions, UDOS currently does not implement Swarm quorum abort or drift baseline detection, and their impact on completion rate/mis-isolation rate is all [RESULT NEEDED]**. This paper also takes "honest failure" as a first-class result—a system that cannot roll back or cascade-suppress, even if it can complete, is not an honest fail-safe system.

Keywords: fail-safe; recovery; Swarm quorum abort; behavioral drift; automatic rollback; cascade containment; kill-switch

Structured Abstract (Background problem → Argument → Evidence → Contributions)

- **Background problem:** P3 can break circuits node-by-node, roll back snapshots, and honestly fail via kill-switch, but its "failure stop" is single-node rule-triggered, lacking three things: quorum abort when the swarm is led astray together, drift detection and rollback without injected faults, and cascade suppression when faults spread across nodes.
 - **Core argument:** Fail-safe and recovery should not only react to "known fault types" but make first-class defenses against three **failure forms not covered by P3**: "(a) swarm-level joint induction, (b) drift without injection, (c) cross-node fault spread"; these three correspond respectively to Pillar 3's quorum abort, drift baseline rollback, and cascade containment.
 - **Evidence threads:** (1) the single-seed pilot of UDOS P3 three-level governance (cpu-proto, 24/24 completion, 3 nodes isolated, 5 reworks, honest kill-switch in the extreme scenario); (2) the F3.3 / F3.4 framework points of AI SAFE² Pillar 3 (official-website scope, thresholds are its claims); (3) the primary-metric and guardrail design of P3's pre-registered RCT (design first).
 - **Contributions:** (i) extending P3's node-by-node governance into a three-layer recovery structure of swarm-level quorum abort + drift baseline rollback + cascade containment; (ii) connecting the three things respectively to UDOS's existing components (committee mechanism, hash-chain snapshots, TransferBundle chain) rather than starting anew; (iii) listing the pending [RESULT NEEDED] list and falsifiable conditions.
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1 What P3 Has Closed-Looped, and What It Still Lacks

1.1 P3's Closed Loop

P3's governance goal is defined as three things: **recover when recovery is possible, isolate and reassign when recovery is impossible, and honestly stop and leave a complete audit trail when utterly impossible**. It measures these three things with the discipline of a pre-registered RCT: the primary metric completion rate = accepted/total, and guardrails are rework, messages, mis-isolation rate, Trace break count, and kill-switch mis-trigger rate. In the cpu-proto single-seed pilot, this closed loop behaves stably when "known faults are injected"—24/24 completion, all 3 faulty nodes isolated, 5 reworks, and honest kill-switch in the extreme scenario.

1.2 Three Failure Forms Not Covered

P3's fault model is **human-injected, of known type, node-by-node** (crash / drop_context / byzantine one each). It does not answer three more hidden failure classes:

1. **Swarm-level joint induction:** It is not one node broken but all nodes led astray by the same wrong context—P3 looks node-by-node at "whether each node is normal," and at this point each node's circuit breaker shows normal, but the whole swarm is converging in the wrong direction. P3's kill-switch is a single-point rule, with no mechanism for "a group of healthy Agents jointly judging whether the swarm has drifted."
2. **Drift without injection:** The system has no fault injected, but its behavior distribution quietly departs from the historical healthy baseline over multiple rounds (such as the reassignment ratio quietly rising and the context-loss rate in handoffs slowly climbing). P3 only reacts to "fault events," not to "gradual changes in behavior distribution."
3. **Cascade spread:** A node isolated by circuit breaking may pass wrong context to downstream through the handoff chain, making downstream nodes gradually abnormal as well. P3's node-by-node circuit breaking isolates the source but does not suppress influence spreading along the Trace path.

Figure 6 Three hidden failure forms not covered by P3

Figure 6 Three hidden failure forms—swarm joint induction, drift without injection, and fault cascade spread—and their corresponding extensions. Conceptual illustration.

These three failure forms are precisely the objects systematized by the external AI SAFE² framework Pillar 3 (Fail-Safe & Recovery).

Figure 5 P3 node-by-node governance to Pillar 3 three-layer recovery

Figure 5 Left: the node-by-node circuit breaking/snapshot/reassignment/kill-switch P3 has implemented; right: the Swarm quorum abort, drift baseline rollback, and cascade containment Pillar 3 extends. Conceptual illustration, not measured data.

2 Direction One: Swarm Quorum Abort (F3.3)

2.1 From "Single-Point Rule Ignition" to "Healthy Majority Judgment"

F3.3 (Swarm Quorum Abort, official-website scope) of AI SAFE² Pillar 3 claims: when **over 25% of Agents become abnormal, a coordinated shutdown should be triggered, and cryptographic attestation**

weighting is used to resist Sybil attacks—i.e. to prevent a few nodes spoofing many identities from faking an "all is well" signal.

Map it onto UDOS: P3's global kill-switch is currently a single-point rule of "trigger when any guardrail threshold is crossed." Swarm quorum abort upgrades it into **the majority judgment of a group of healthy Agents**—this shares a source with the committee mechanism of P16/P26:

\$\$ \text{\text{swarm_abort}} = \text{\text{true}}, \quad \text{\text{if the abnormal-Agent share } } \rho \geq \rho_0 \text{\text{ and the healthy committee's attestation majority supports shutdown}} \$\$

(ρ_0 is a design threshold, the framework claims about 25%, not reproduced in UDOS.) The key difference: the single-point rule only looks at metrics, while Swarm quorum abort looks at **"whether a group of unsuspected Agents independently think the swarm has drifted."** The latter can recognize the case of "all metrics normal but direction wrong"—which is precisely what P3's node-by-node model cannot see.

2.2 Sybil Resistance

F3.3's attestation weighting corresponds to the "credibility signal" discussed in P26: weighted shutdown votes cannot be faked by a few spoofed identities. UDOS's stance is consistent with P26—weights are computed with rule predicates (historical health), not dynamically estimated with learned models, and it is not implemented before the steelman contrast runs through. Swarm quorum abort is currently a design proposal, [RESULT NEEDED].

3 Direction Two: Behavioral Drift Baseline Automatic Rollback (F3.4)

3.1 From "Reacting to Fault Events" to "Reacting to Behavior Distribution"

F3.4 (Behavioral Drift Baseline & Rollback, official-website scope) of AI SAFE² Pillar 3 claims: build a **baseline** for the system's normal behavior, and when running behavior departs from the baseline beyond a threshold, **automatically roll back** to the last known healthy snapshot.

This precisely fills P3's second gap—gradual change without injection. P3's Trace hash chain is already recording each step's state; what F3.4 does is add a layer of **distribution monitoring** on this chain:

- Baseline: build a distribution with key behavior metrics from historical healthy runs (reassignment ratio, context-loss rate in handoffs, rework rate, average fan-in);
- Drift: the current window's metrics depart from the baseline distribution beyond a threshold;
- Rollback: roll back along the Trace hash chain to the last non-drifted snapshot, rather than waiting for a fault event to trigger.

The benefit of this design is **reusing UDOS's existing assets**: snapshots are already in P3, the rollback path is already in the hash chain, and what is added is only "distribution monitoring + automatic rollback trigger." It is an increment to P3's pre-registered RCT—extending guardrails from "event counts" to "distribution departure."

3.2 Connection with P3's Extreme Scenario

P3 in the "all-domain faults + zero retry budget" extreme scenario **honestly triggers kill-switch and reports non-completion**—this is where it is honest. F3.4's drift rollback is not meant to replace this honest failure, but to intervene at an earlier stage of "not yet extreme but already drifting," letting the system roll back **before** entering the extreme scenario and reducing the number of times it reaches kill-

switch. It does not contradict "honest failure": after rollback fails, kill-switch still provides the bottom line.

4 Direction Three: Cascade Containment

4.1 Suppressing Influence Spread Along the Trace Path

P3's node-by-node circuit breaking isolates the fault source, but faults may jump downstream through the handoff chain. Cascade containment requires: while isolating the faulty node, **mark the affected downstream nodes along its Trace / TransferBundle path**, giving special handling to these nodes' in-flight work orders (down-weighting, re-checking, or temporary isolation), rather than letting them keep working with polluted context.

This direction reuses P10's TransferBundle (P10 has implemented the six-field handoff contract, 26 contract tests verified): the handoff chain itself is a "who passed to whom" graph, and cascade containment only needs to do **bounded-radius influence-spread marking** from the fault source on this graph, with nodes outside the radius unaffected—avoiding the overreaction of "one broken, all stop."

5 The Three-Layer Recovery Structure and Its Relationship with P3

Table 1 Extension mapping from P3 node-by-node governance to Pillar 3 three-layer recovery

Failure form	P3 current response	Pillar 3 extension	Reused UDOS component	Status
Single-node crash/drop	AgentBreaker three-level circuit breaking	—	<code>circuit_breaker.py</code>	implemented (cpu-proto)
Byzantine	QA committee + quality-failure accumulation	—	<code>consensus.py</code>	implemented (cpu-proto)
Swarm joint induction	single-point kill-switch rule	F3.3 Swarm quorum abort + attestation weighting	P16/P26 committee mechanism	design proposal, not implemented
Drift without injection	none (only reacts to events)	F3.4 drift baseline automatic rollback	P3 hash-chain Trace snapshots	design proposal, not implemented
Fault cascade spread	node-by-node isolation	bounded-radius containment along the TransferBundle path	P10 handoff chain graph	design proposal, not implemented
Utterly unrecoverable	kill-switch honest failure	not replaced, as bottom line	<code>governance.py</code>	implemented (cpu-proto)

The judgment of Table 1 is clear: **P3 makes the fault governance of "single-node, known-type" into a measurable closed loop; what Pillar 3 fills are the three more hidden failure forms of "swarm-level, gradual, cascade."** All three can connect to UDOS's existing components without starting anew—this is why they can be registered as "design extensions" rather than "brand-new systems."

6 Dross / Failure Boundaries / Falsifiable Conditions

Dross (what not to do):

1. **Do not write Swarm quorum abort as "a more sensitive kill-switch."** Its value is not igniting more frequently but recognizing "metrics normal but direction wrong"—if after implementation it only lowers the threshold, it degenerates into a more aggressive kill-switch, losing the meaning of healthy majority judgment.
2. **Do not let drift rollback replace honest failure.** Rollback intervenes in the early drift stage; when the system has already entered P3's extreme scenario, the correct behavior is still kill-switch honestly reporting non-completion, not infinite rollback.
3. **Do not make cascade containment into "one broken, all stop."** Influence spread must have a bounded radius, with no reaction outside the radius; unbounded containment amounts to magnifying a single fault into a full-matrix shutdown.

Failure boundaries:

- Swarm quorum abort's attestation weighting still faces the "weight-estimation manipulation" problem discussed in P26—healthy Agents themselves may be induced, and quorum abort cannot distinguish "the healthy majority really drifting" from "the healthy majority being colluded";
- The drift baseline has no defense against "fault types that have never appeared"—the baseline only knows "departure from the known healthy distribution," not "whether the departure is a brand-new attack";
- Cascade containment's bounded radius is a prior parameter: too small a radius misses real downstream pollution; too large a radius over-isolates.

Falsifiable conditions:

- If Swarm quorum abort in a "swarm joint induction" injection experiment neither lowers the mis-trigger rate nor the miss rate compared to P3's single-point kill-switch, the benefit hypothesis of the F3.3 extension is overturned;
- If drift baseline rollback's mis-rollback rate on fault-free normal runs is significantly higher than the number of drift events it captures, F3.4's side effects outweigh its benefits;
- If cascade containment's downstream mis-isolation rate on injected cross-node pollution is not significantly lower than node-by-node isolation, F3 direction three has no engineering value in the UDOS scenario.

7 Conclusion

v7.5.0 P3 makes fault governance into a measurable closed loop that "can recover, can isolate, can honestly fail"; AI SAFE² Pillar 3 reminds us this closed loop still lacks three layers—swarm-level quorum abort, drift baseline automatic rollback, cascade containment. This paper's stance is in the same line as P3: **first register these three things as design extensions, connecting to existing components, rather than immediately implementing.** The reason is the same as P26—the reason P3's pre-registered RCT is credible is that its governance mechanisms are independently switchable and causally measurable; any new mechanism (quorum abort, drift detection, cascade containment) must first become an "independently switchable, marginally measurable" factor before entering the primary-metric experiment.

A system that cannot recognize the swarm jointly drifting, cannot automatically roll back when quietly drifting, and cannot suppress fault spread along the handoff chain—even if it can complete node-by-node—is not an honest fail-safe system. But before putting these mechanisms into safety-critical paths, it must first prove they are not "more sensitively mis-triggering" under a steelman contrast—consistent with P3's usual restraint.

References

Classical literature

1. Castro, M., & Liskov, B. (1999). Practical Byzantine Fault Tolerance. OSDI 1999. (the classical skeleton of fault tolerance and committee majority judgment)
2. Microservice circuit breaker pattern (circuit breaker: closed/open/half-open, error-rate threshold, cooldown probe). [CITATION NEEDED: the original source of the microservice circuit breaker pattern pending verification]

2026 frontier materials (existence verified; thresholds are framework claims, not reproduced in UDOS)

1. Cyber Strategy Institute (2026). AI SAFE² governance framework v3.0 (2026-05-18), **Pillar 3 Fail-Safe & Recovery**: F3.3 Swarm Quorum Abort (coordinated shutdown at 25%+ abnormal, cryptographic attestation weighting against Sybil), F3.4 Behavioral Drift Baseline & Rollback (automatic rollback on behavioral drift baseline), circuit breaker/circuit breaking/cascade containment.

UDOS paper volume (evidence grades marked in text)

1. UDOS v7.5.0 P3 "Causal Effect of Fault-Breaker Governance on Agent Matrix Completion Rate" (three-level circuit breaking / snapshot rollback / reassignment + unique Owner / hash-chain Trace / explicit Stop Condition; primary metric completion rate accepted/total; 24 work orders, 9 specialists, 7 QA including 2 Byzantine, cpu-proto single-seed pilot).
 2. UDOS v7.5.0 P10 "AgentAsTool and the Measurability of TransferBundle Handoff Information Loss" (six-field handoff contract, 26 contract tests verified).
 3. UDOS v7.6.0 P16 "Typed Finality", P26 "Weighted BFT and Reputation" (the mechanism source of Swarm quorum abort and attestation weighting).
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Evidence Discipline and Reproduction Notes

- This paper is a **design extension** paper with no new experiment; UDOS's current `udos7/topology/circuit_breaker.py`, `governance.py`, and `consensus.py` only implement P3's node-by-node governance and **do not implement** Swarm quorum abort, drift baseline detection, or cascade containment.
- P3's existing numbers are cpu-proto single-seed pilot point estimates (24/24 completion, 3 nodes isolated, 5 reworks); "all-domain faults + zero retry budget triggering kill-switch" is an extreme scenario constructed by design, whose exact accepted count and governance-item details await re-running (P3 text marks [RESULT NEEDED]) and are not within this clean/stressed two runs.
- The 25% threshold, attestation weighting, and drift baseline of AI SAFE² Pillar 3 are **framework claims, official-website scope**, not reproduced in UDOS; this paper only does mechanism mapping and does not claim these thresholds are optimal in the UDOS scenario.

- The impact of the three extensions on completion rate/mis-isolation rate/mis-trigger rate is all [RESULT NEEDED] ; if implemented, they must, per P3's pre-registered RCT discipline, become independently switchable factors, with ≥ 30 seeds before submission.